AE4868-II (2025/2026) - Assignment 2

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Github link: ... *this is my link*

Time spent per exercise:

1. 8 hours
2. 13 hours
3. 24 hours
4. 12 hours

People you worked with: Valentina Balzanelli 6553761

Question 1

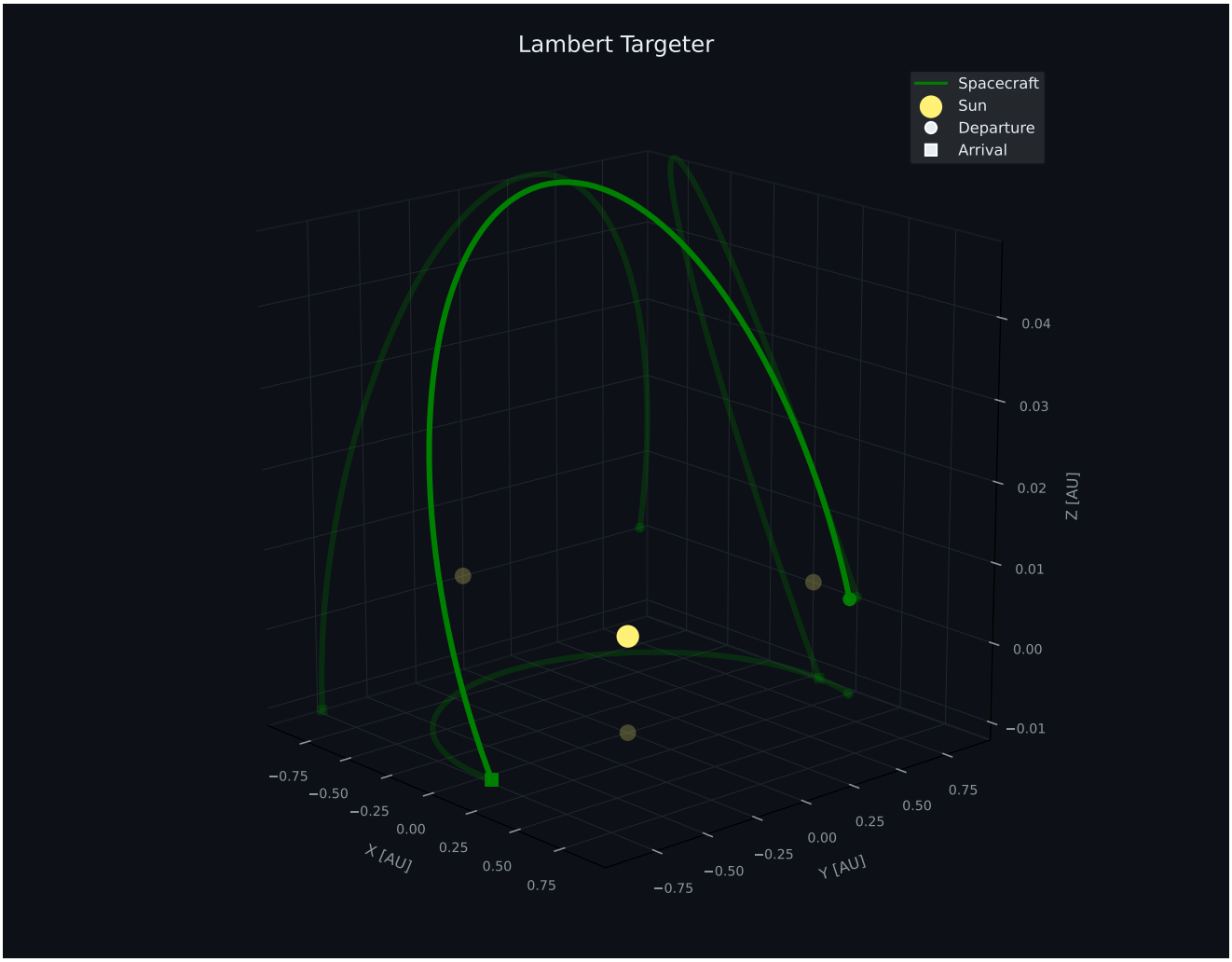

Figure 1: 3D plot of the unperturbed propagated trajectory

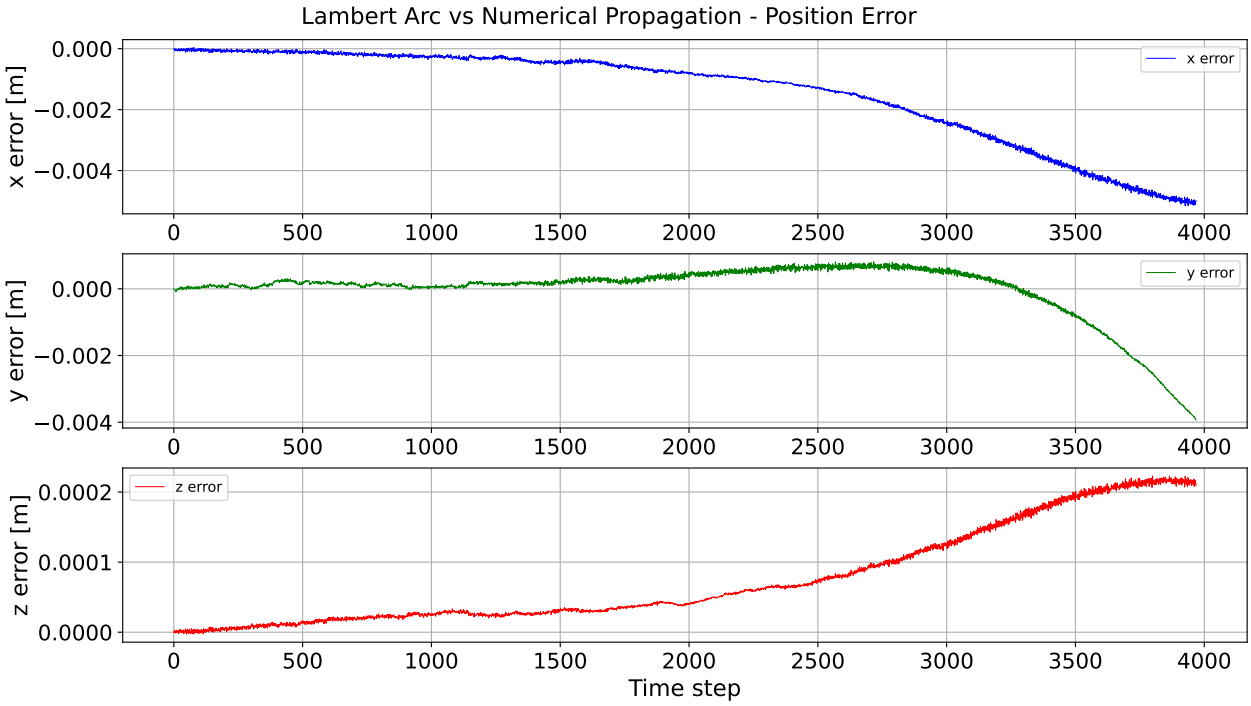

Figure 2: Magnitude of positional propagation errors in the three directions between the analytical solution of the Lambert Arc and the numerically propagated one.

List of assumptions

1. *Heliocentric two-body, point-mass Sun.* Lambert assumes $\ddot{\mathbf{r}} = -\mu_\odot \mathbf{r}/r^3$, with constant $\mu_\odot$, no oblateness, relativity, or time-varying gravity. This is **TRUE**:
   the unperturbed-propagation uses only Sun point-mass gravity.

2. *No additional accelerations.* Lambert neglects third-body, SRP, drag, thrust, and any empirical acceleration. This is **TRUE**:
   the propagated arc is purely ballistic and unperturbed.

3. *Sun-centered state formulation.* Lambert is written for the spacecraft state relative to the central body. This is **TRUE**:
   both the Lambert arc and the numerical propagation are heliocentric, so the Sun's motion with respect to the SSB is not a relevant discrepancy.

4. *Fixed endpoints and time of flight.* Lambert assumes $\mathbf{r}_1$, $\mathbf{r}_2$, and Δt to be prescribed. This is **PARTIALLY TRUE**:
   the same departure state and final epoch are imposed, but the propagated final state is only approximately recovered because the numerical propagation solves an initial-value problem.

5. *Exact Keplerian conic arc.* Lambert gives one exact continuous conic joining the endpoints. This is **PARTIALLY TRUE**:
   the propagated dynamics are the same, but the RK4 integration introduces truncation and round-off error, producing small oscillations and a weak drift; this is not relevant here because the mismatch remains extremely small.

Question 2

a

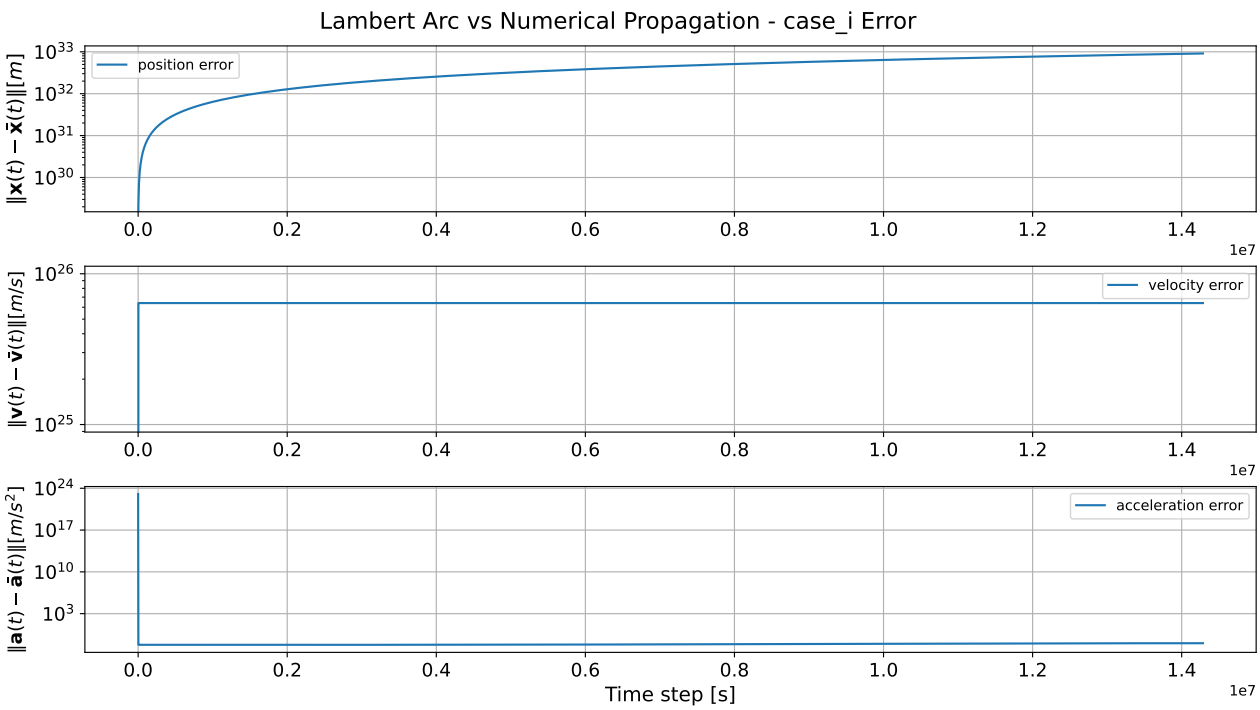

Figure 3

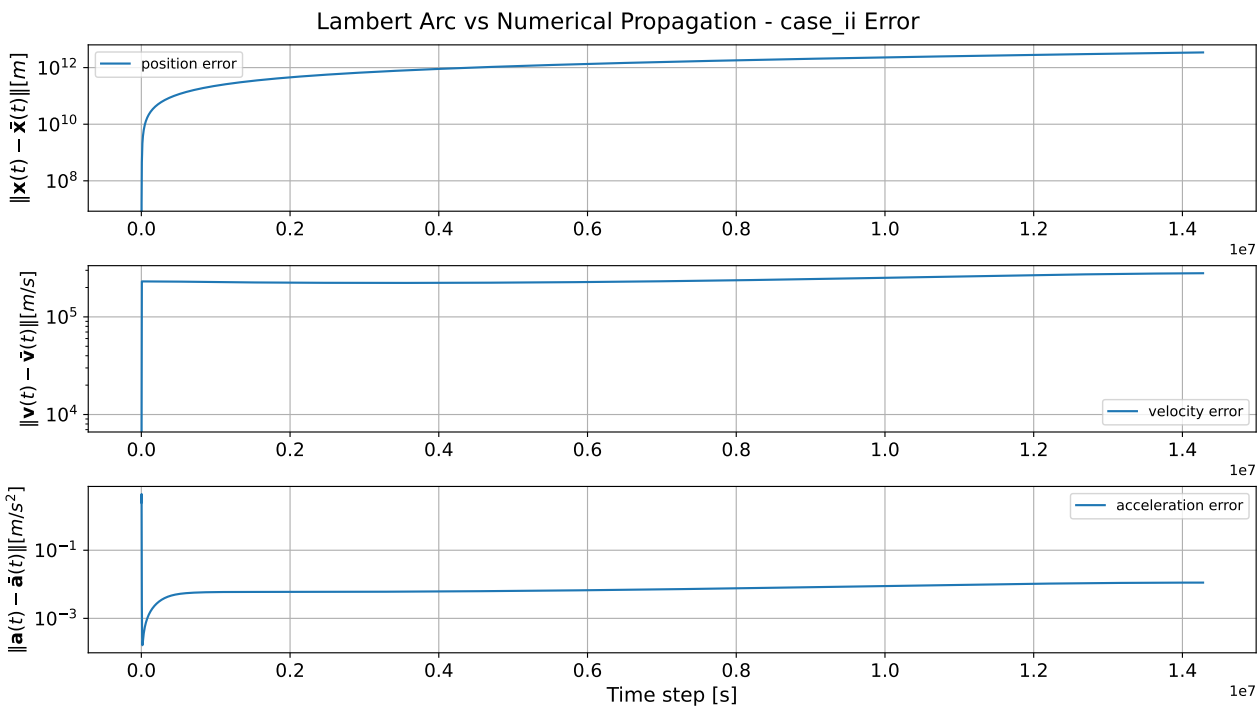

Figure 4

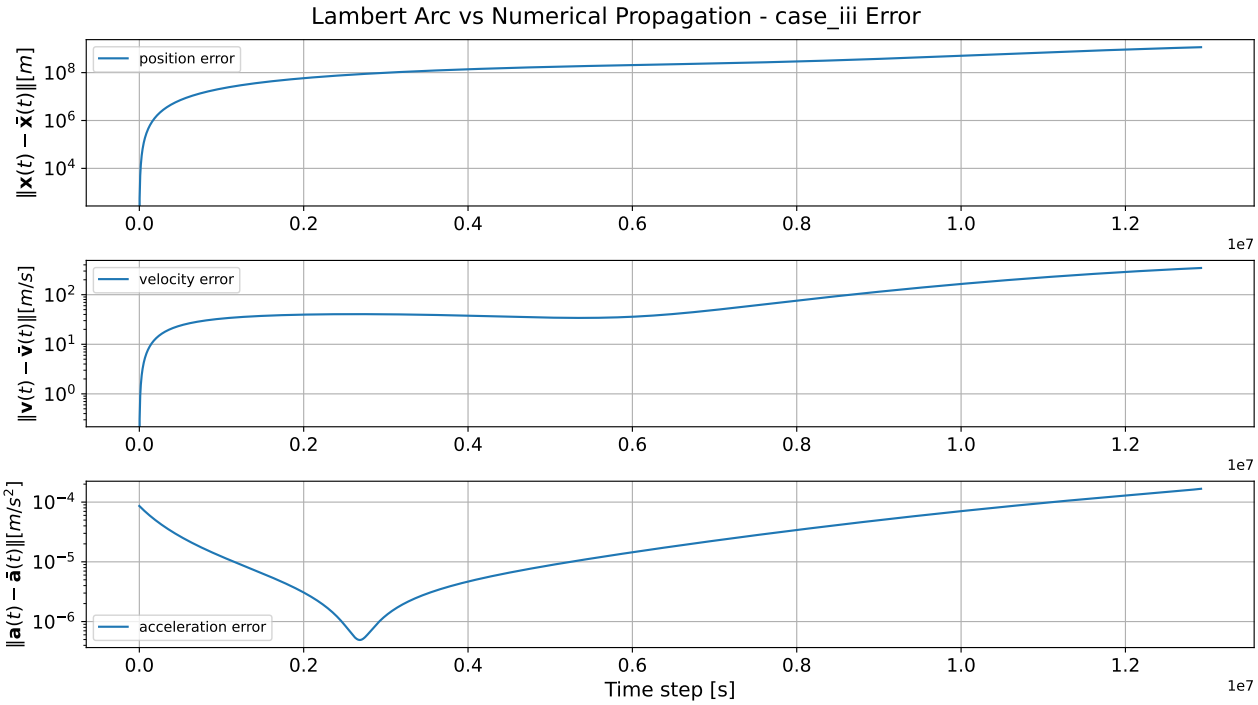

Figure 5

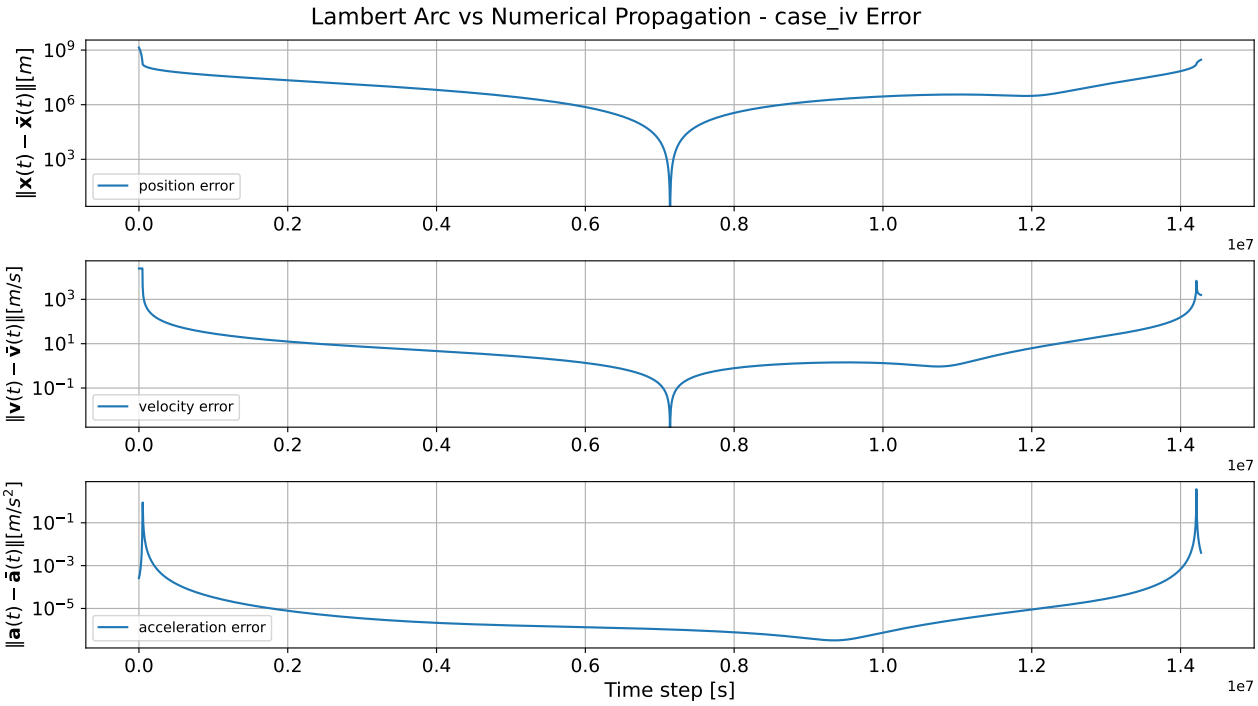

Figure 6

b

The Lambert Arc assumes purely heliocentric two-body motion, whereas near departure and arrival planetary gravity remains non-negligible, especially inside the SOI, producing an acceleration mismatch that propagates into a drift in Δr. Increasing the buffer time moves the boundaries to regions where the solar term dominates. In case iv, starting at t_{mid} places the initial condition in a weak-perturbation region and halves the integration length to each endpoint. Both of these reasons reduce the accumulated drift, since perturbations are strongest near the SOI boundaries and drift grows with integration length.

c

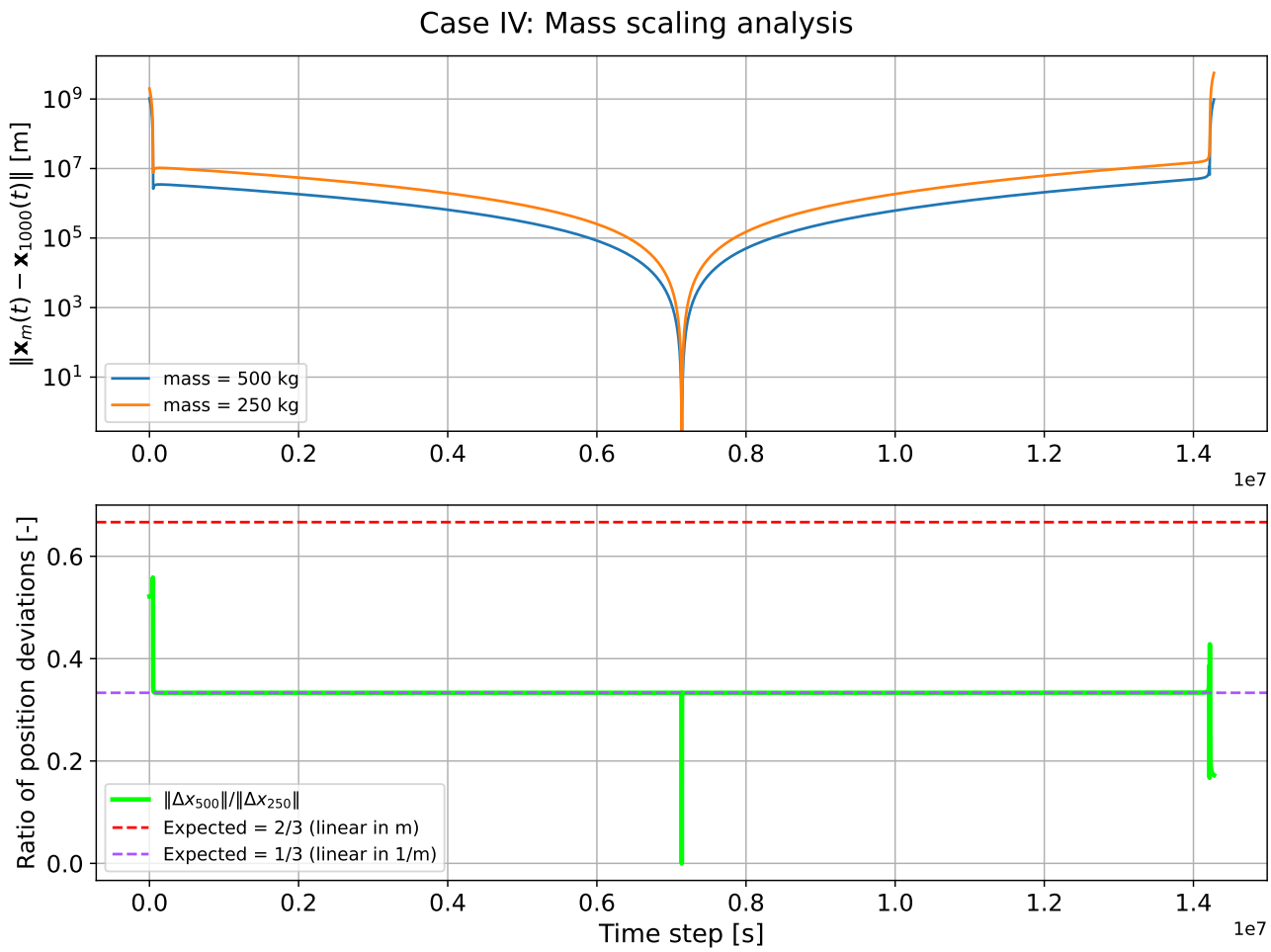

Figure 7: Top: position difference w.r.t. case iv at 1000 kg. Bottom: Figure of Merit $\mathcal{F}(t) = \|\Delta\mathbf{x}_{500}\|/\|\Delta\mathbf{x}_{250}\|$.

The figure of merit is defined as the ratio of position deviations with respect to the 1000 kg reference at each timestep:

$$\mathcal{F}(t) = \frac{\|\mathbf{x}_{500}(t) - \mathbf{x}_{1000}(t)\|}{\|\mathbf{x}_{250}(t) - \mathbf{x}_{1000}(t)\|} \tag{1}$$

Since the SRP acceleration scales as $1/m$ (constant force, different mass), the position deviation accumulates as $\Delta\mathbf{x} \propto (1/m - 1/m_{ref})$. Two analytical predictions follow:

$$\mathcal{F}_{\text{linear}} = \frac{500 - 1000}{250 - 1000} = \frac{2}{3}, \qquad \mathcal{F}_{\text{inverse}} = \frac{1/500 - 1/1000}{1/250 - 1/1000} = \frac{1}{3} \tag{2}$$

The figure confirms $\mathcal{F}(t) \approx 1/3$ throughout, thus the mass effect is inversely proportional to m. This holds while $\|\Delta\mathbf{x}\|/r \ll 1$, so while the coupling between the perturbed position and gravitational acceleration remains negligible.

Question 3

a

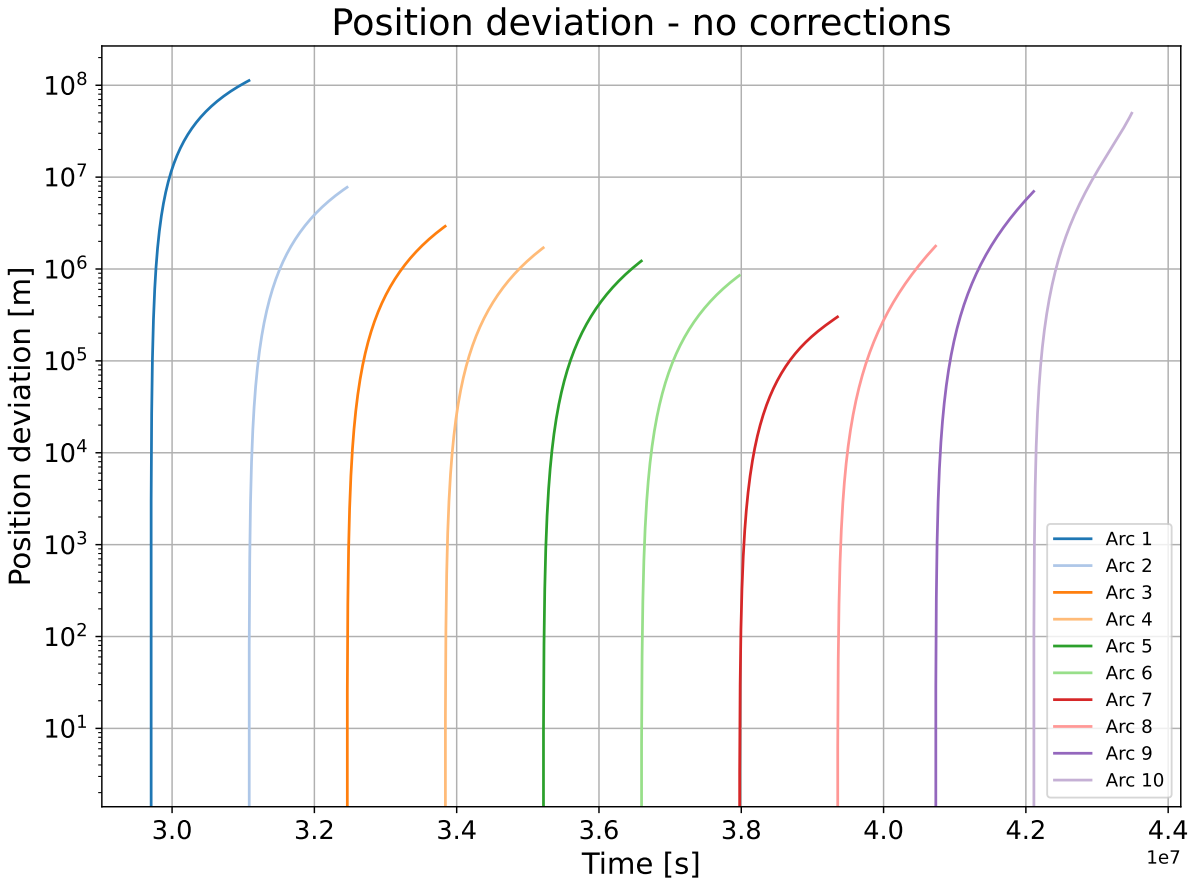

Figure 8

b

The trajectory is divided into N arcs, where arc i spans from t_i to t_{i+1}, with $t_1 = t_0$ and $t_{N+1} = t_E$. Starting from the linearised propagation equation for arc i:

$$\begin{pmatrix} \Delta\mathbf{r}_i(t_{i+1}) \\ \Delta\mathbf{v}_i(t_{i+1}) \end{pmatrix} = \Phi_i(t_{i+1}, t_i) \begin{pmatrix} \Delta\mathbf{r}_i(t_i) \\ \Delta\mathbf{v}_i(t_i) \end{pmatrix} \tag{3}$$

Assumption 1: The initial position of the arc is fixed at the Lambert reference, so $\Delta\mathbf{r}_i(t_i) = \mathbf{0}$. The free parameter vector is the velocity correction $\Delta\mathbf{v}_i \equiv \Delta\mathbf{v}_i(t_i)$ since we are considering impulsive thrusts. It is therefore only useful to retain the upper three rows and the last three columns of Φ_i, defining $\Phi_{i,rv} \equiv \partial\mathbf{r}(t_{i+1})/\partial\mathbf{v}(t_i) \in R^{3\times 3}$, which are the terms which map the imposed initial velocity variation to the final position deviation through the linearization:

$$\Delta\mathbf{r}_i(t_{i+1}) = \Phi_{i,rv}(t_{i+1}, t_i)\, \Delta\mathbf{v}_i \tag{4}$$

Assumption 2: No constraint is placed on $\Delta\mathbf{v}_i(t_{i+1})$, so only the position rows are required.
The correction must cancel the deviation between the unthrusted numerical propagation $\mathbf{r}_i(t_{i+1})$ and the reference $\bar{\mathbf{r}}(t_{i+1})$:

$$\Phi_{i,rv}(t_{i+1}, t_i)\, \Delta\mathbf{v}_i + \big(\mathbf{r}_i(t_{i+1}) - \bar{\mathbf{r}}(t_{i+1})\big) = \mathbf{0} \tag{5}$$

Since $\Phi_{i,rv}$ is a square 3×3 matrix the system is exactly determined and the unique solution is:

$$\boxed{\Delta\mathbf{v}_i = -\Phi_{i,rv}^{-1}(t_{i+1}, t_i)\, \big(\mathbf{r}_i(t_{i+1}) - \bar{\mathbf{r}}(t_{i+1})\big)} \tag{6}$$

c

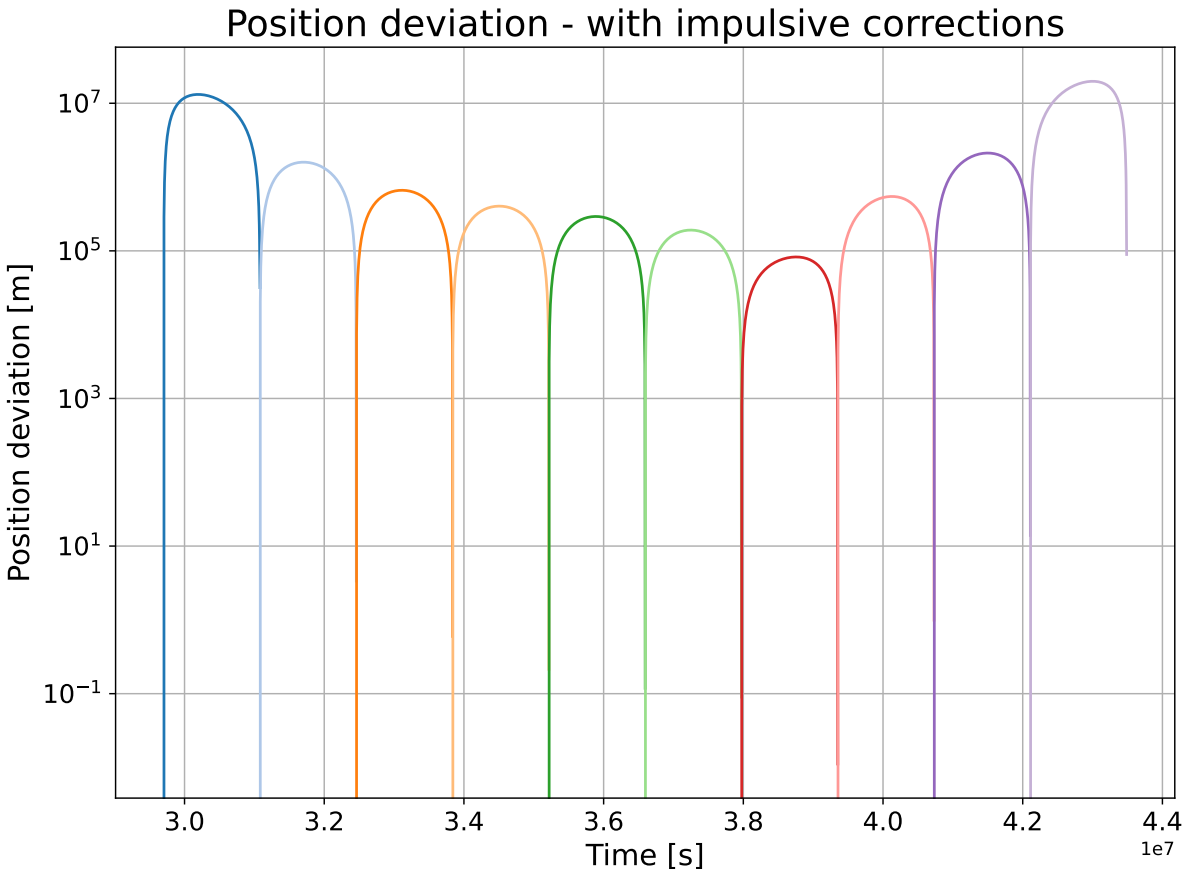

Figure 9: Overview of the arcs with the impulsive corrections applied in correspondance with the Lambertian Arc at the beginning of each sub-arc

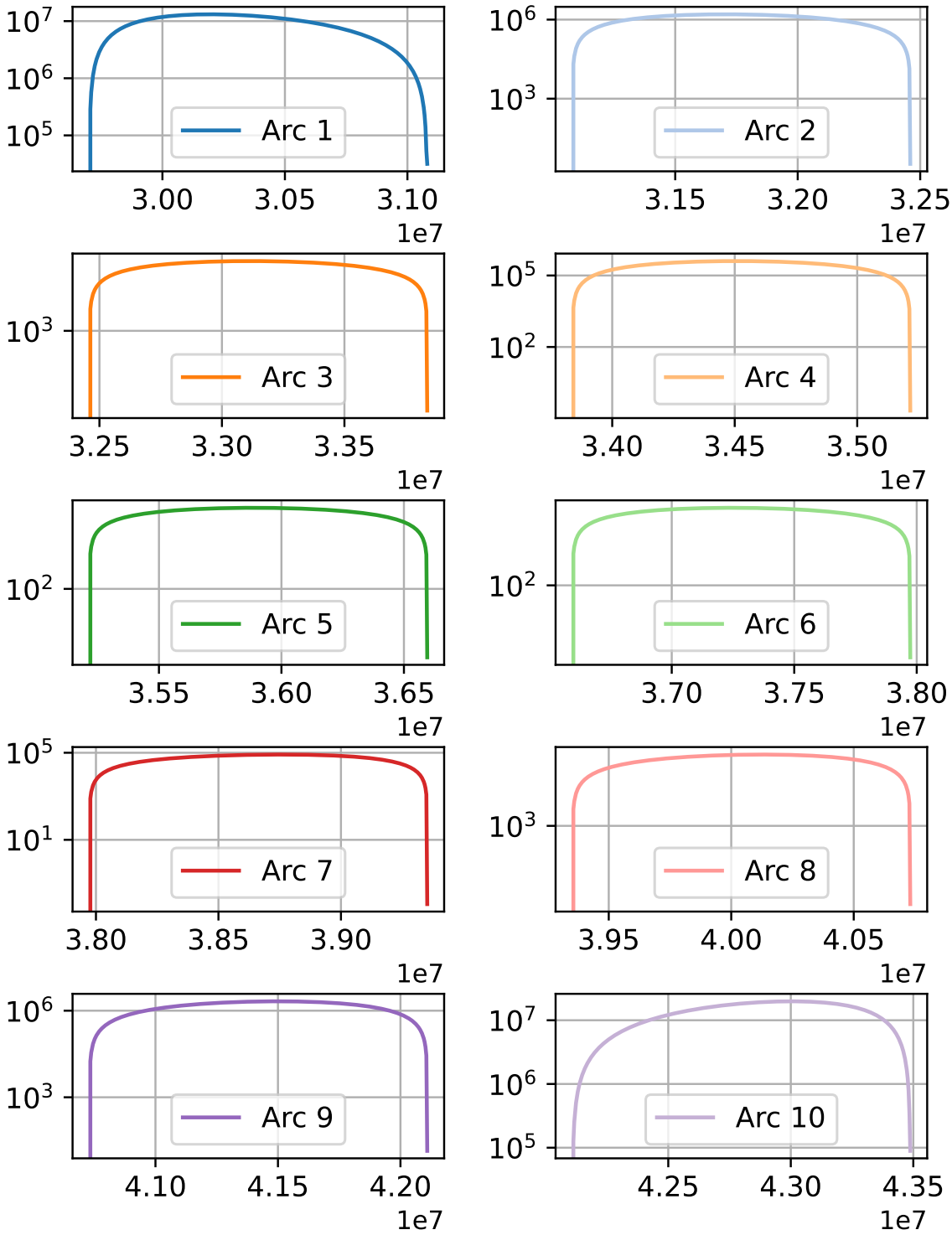

Figure 10: Single arcs plotted to highlight the values on each sub-arc, and their evolution.

d

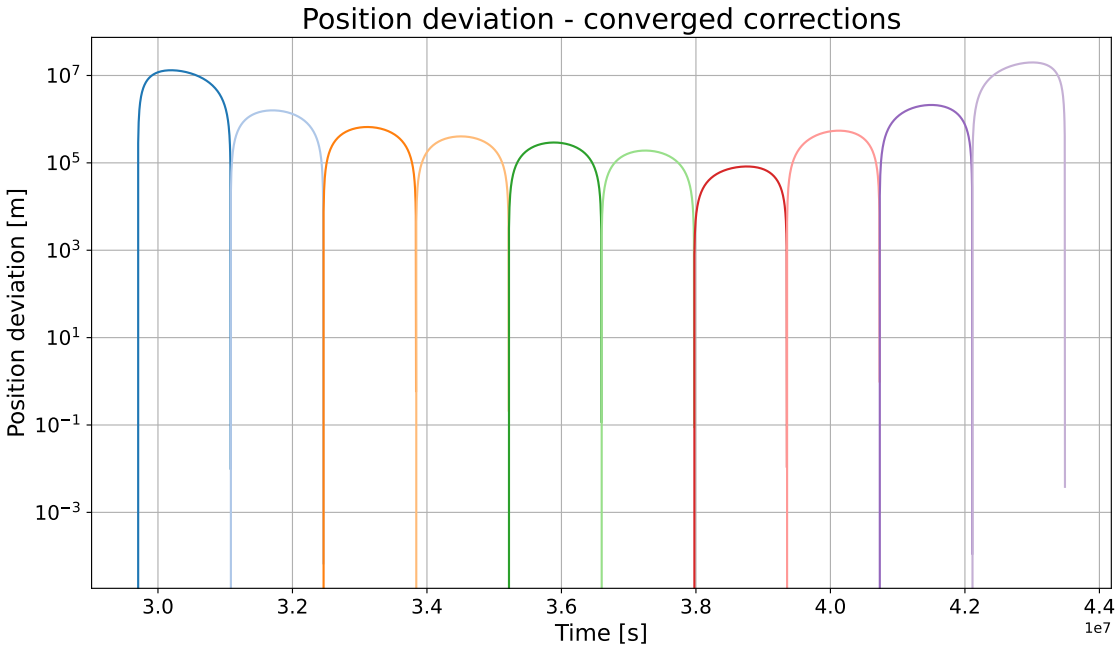

Figure 11: Overview of the arcs with the impulsive iterated and converged corrections applied in correspondance with the Lambertian Arc at the beginning of each sub-arc

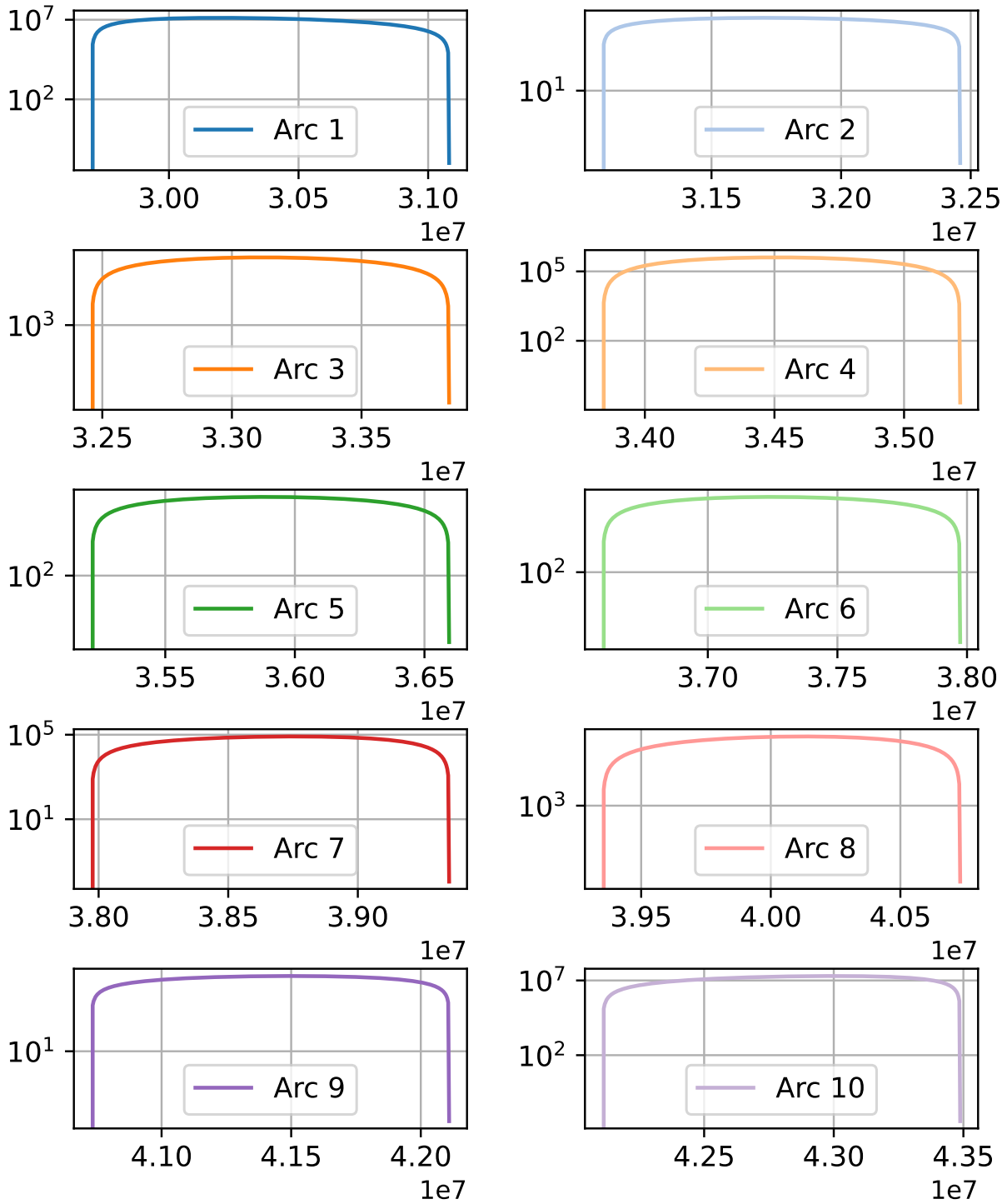

Figure 12: Single arcs plotted to highlight the values on each sub-arc, and their evolution.

The number of corrective iterations required per arc until $\|\mathbf{r}_i(t_{i+1}) - \bar{\mathbf{r}}(t_{i+1})\| < 1\,\mathrm{m}$ is given in Table 1.

Arc i	Propagations required
1	3
2	2
3	1
4	1
5	1
6	1
7	1
8	1
9	2
10	4

Table 1: Number of corrective propagations per arc until convergence to 1 m.

Arcs 1 and 10 border the Earth departure and Venus arrival spheres of influence thus planetary gravity drives the numerical trajectory far from the Lambert reference. Since the State Transition Matrix correction is a linearization valid only in a neighbourhood of the reference, when the deviation is large, the first correction cannot fully cancel the residual, and iterations re-apply the same linear correction to the remaining smaller residual to reach convergence. Intermediate arcs experience weaker perturbations, so a single correction suffices. Arcs 2 and 9 are close enough to the perturbing bodies to require one additional iteration, but their initial deviations are modest enough to converge faster than arcs 1 and 10.

e

Each arc requires a single impulsive manoeuvre $\Delta\mathbf{v}_i$ at its start, accumulated over all corrective iterations. Since the arcs are independent and position mismatches between arcs are ignored, the total ΔV over the full trajectory is:

$$\Delta V = \sum_{i=1}^{N} \|\Delta\mathbf{v}_i\| \tag{7}$$

where $\Delta\mathbf{v}_i = \sum_k \delta\mathbf{v}_i^{(k)}$ is the cumulative velocity correction applied to arc i summed through all iterations k. It is relevant to do the vectorial summation and not in norm due to the triangular inequality. Evaluating over the 10 arcs:

$$\Delta V \approx 133.1 \text{ m/s} \tag{8}$$

Question 4

a

Starting from the linearized propagation equation:

$$\begin{pmatrix} \Delta\mathbf{r}(t_E) \\ \Delta\mathbf{v}(t_E) \end{pmatrix} = \Phi(t_E, t_0) \begin{pmatrix} \Delta\mathbf{r}(t_0) \\ \Delta\mathbf{v}(t_0) \end{pmatrix} + \mathbf{S}(t_E)\,\mathbf{p} \tag{9}$$

Assumption 1: The initial state is held fixed at the reference state on the Lambert Arc, thus $\Delta\mathbf{x}(t_0) = \mathbf{0}$. The state-transition-matrix term therefore vanishes:

$$\begin{pmatrix} \Delta\mathbf{r}(t_E) \\ \Delta\mathbf{v}(t_E) \end{pmatrix} = \mathbf{S}(t_E)\,\mathbf{p} \tag{10}$$

Assumption 2: No constraint is placed on the final velocity difference $\Delta\mathbf{v}(t_E)$. It is therefore only useful to retain the upper three rows of $\mathbf{S}(t_E)$, defining $\mathbf{S}_r(t_E) \in R^{3\times 3}$ as the rows corresponding to the position change:

$$\Delta\mathbf{r}(t_E) = \mathbf{S}_r(t_E)\,\mathbf{p} \tag{11}$$

The thrust must cancel the deviation between the unthrusted numerical propagation $\mathbf{r}_0(t_E)$ and the reference $\bar{\mathbf{r}}(t_E)$ and this can be achieved by imposing the following:

$$\big(\mathbf{r}_0(t_E) - \bar{\mathbf{r}}(t_E)\big) + \mathbf{S}_r(t_E)\,\mathbf{p} = \mathbf{0} \tag{12}$$

Since $\mathbf{S}_r(t_E)$ is a square 3×3 matrix the system is exactly determined and the unique solution is:

$$\boxed{\mathbf{p} = -\big[\mathbf{S}_r(t_E)\big]^{-1}\big(\mathbf{r}(t_E) - \bar{\mathbf{r}}(t_E)\big)} \tag{13}$$

b

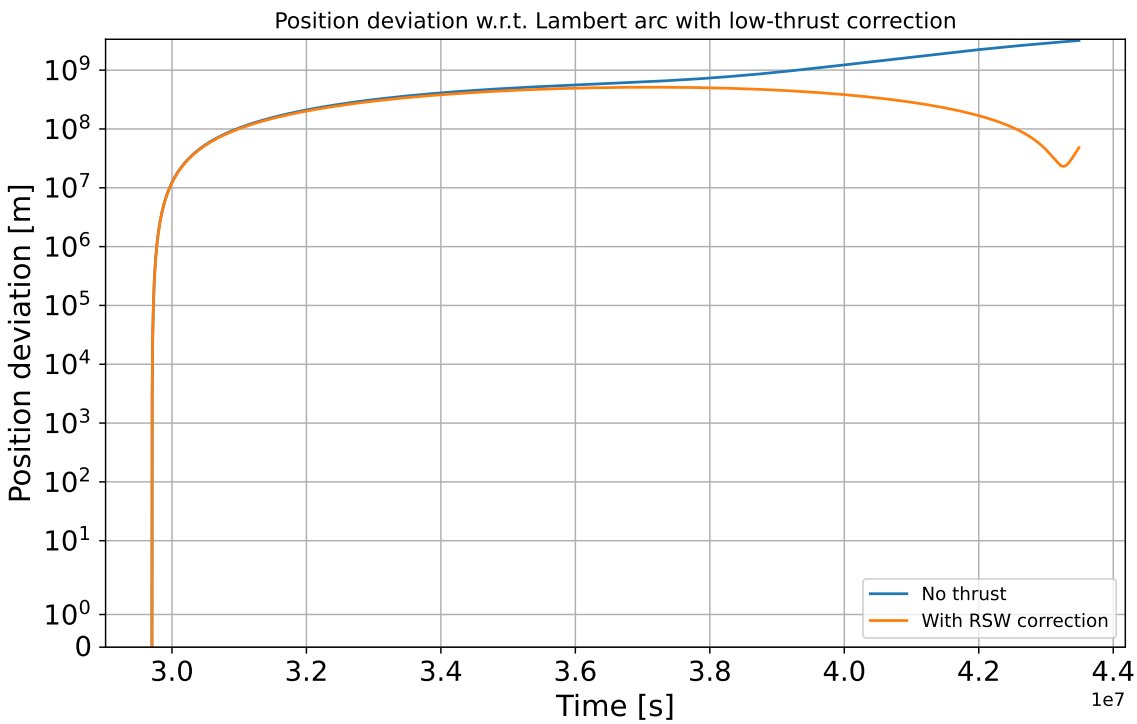

Figure 13

The derived model is valid: **Can be, but not really**
Figure 13 shows that the RSW correction reduces the final position deviation from $\sim 10^9$ m to $\sim 10^7$ m in a single application. This reduction of two orders of magnitude confirms that $\mathbf{S}_r$ correctly captures the dominant sensitivity

of the final position to the applied thrust but the model cannot be considered qualitatively valid. This is because the residual of $\sim 10^7$ m is too large in absolute terms, thus the linearization error is non-negligible. This is expected because the full-arc deviation is large enough that the trajectory departs significantly from the Lambert reference, where the linearization is not too accurate. The model is therefore useful because it provides a strong first correction but iterations are necessary to reach an operationally acceptable residual.

c.i

Starting from the linearized propagation equation for arc 2:

$$\begin{pmatrix} \Delta\mathbf{r}_2(t_E) \\ \Delta\mathbf{v}_2(t_E) \end{pmatrix} = \Phi_2(t_E, t_{mid}) \begin{pmatrix} \Delta\mathbf{r}_2(t_{mid}) \\ \Delta\mathbf{v}_2(t_{mid}) \end{pmatrix} + \mathbf{S}_2(t_E)\,\mathbf{p}_2 \tag{14}$$

Where t_{mid} is the initial propagation time of arc 2.
Assumption 1: Unlike in 4a, the initial state of arc 2 deviates from the Lambert reference, since it is the final state of the previous arc. $\Delta\mathbf{x}_2(t_{mid}) \neq \mathbf{0}$ and the state-transition-matrix term is therefore retained. It is however assumed that the deviated point at t_{mid} remains in a neighbourhood of the Lambert arc, where the linearization is valid.
Assumption 2: As in 4a, no constraint is placed on $\Delta\mathbf{v}_2(t_E)$. It is therefore only useful to retain the upper three rows of Φ_2 and $\mathbf{S}_2$, defining $\Phi_{2,r} \in R^{3\times 6}$ and $\mathbf{S}_{2,r} \in R^{3\times 3}$:

$$\Delta\mathbf{r}_2(t_E) = \Phi_{2,r}(t_E, t_{mid})\,\Delta\mathbf{x}_2(t_{mid}) + \mathbf{S}_{2,r}(t_E)\,\mathbf{p}_2 \tag{15}$$

Setting $\Delta\mathbf{r}_2(t_E) = \mathbf{0}$ corrects for the deviation inherited from arc 1; any residual position error due to perturbations acting during arc 2 is handled by subsequent iterations. The system is exactly determined and the unique solution is:

$$\boxed{\mathbf{p}_2 = -\mathbf{S}_{2,r}^{-1}(t_E)\,\Phi_{2,r}(t_E, t_{mid})\,\Delta\mathbf{x}_2(t_{mid})} \tag{16}$$

Below is the programmatic procedure:

- $\mathbf{p}_1 = \mathbf{p}^{(b)}$ (as required by the assignment)
- I propagate arc 1 from t_0 to t_{mid} with $\mathbf{p}_1$ and compute $\Delta\mathbf{x}_2(t_{mid}) = \mathbf{x}_{arc1}(t_{mid}) - \bar{\mathbf{x}}(t_{mid})$
- I propagate the variational equations for arc 2 along the Lambert reference from t_{mid} to t_E with RSW acceleration enabled, obtaining $\Phi_{2,r}$ and $\mathbf{S}_{2,r}$
- I compute $\mathbf{p}_2 = -\mathbf{S}_{2,r}^{-1}\,\Phi_{2,r}\,\Delta\mathbf{x}_2(t_{mid})$
- I propagate arc 2 from $\mathbf{x}_{arc1}(t_{mid})$ with $\mathbf{p}_2$ applied and check the residual $|\mathbf{r}_2(t_E) - \bar{\mathbf{r}}(t_E)|$
- I repeat the last two steps until the residual falls below the 1 m convergence threshold; at each iteration I compute the correction increment $\delta\mathbf{p}_2 = -\mathbf{S}_{2,r}^{-1}\left(\mathbf{r}_2(t_E) - \bar{\mathbf{r}}(t_E)\right)$ and update $\mathbf{p}_2 \leftarrow \mathbf{p}_2 + \delta\mathbf{p}_2$

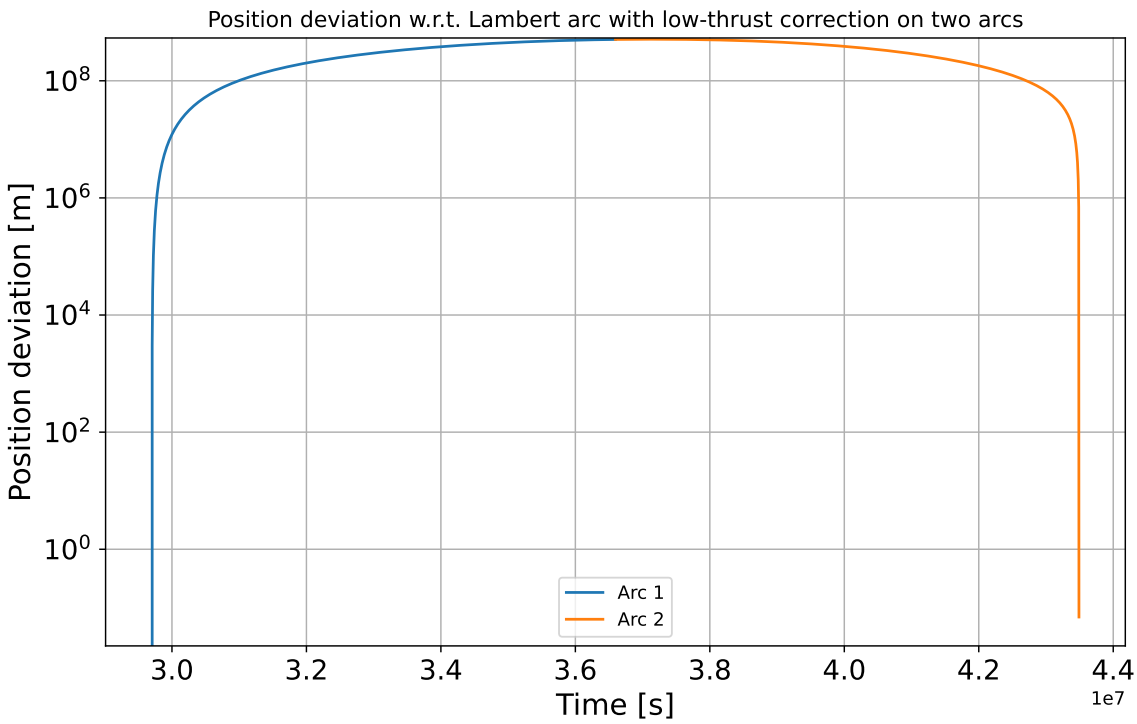

Figure 14

c.ii

The value of p_2 is $1.1852 \cdot 10^{-5}\,\mathrm{m/s^2}$, whereas $p_1 = p^{(b)} = 1.3066 \cdot 10^{-5}\,\mathrm{m/s^2}$, so $p_2 < p_1$ but remains of the same order. The difference is not substantial because the overall transfer to be corrected remains essentially the same. Splitting the trajectory into two continuous-thrust arcs improves the distribution of the control effort, but the first arc already places the spacecraft on an intermediate trajectory designed to approach the Lambert reference in a certain way. The second arc therefore mainly refines the correction, reducing the linearization error without, but cannot drastically change the required thrust level.

d.i

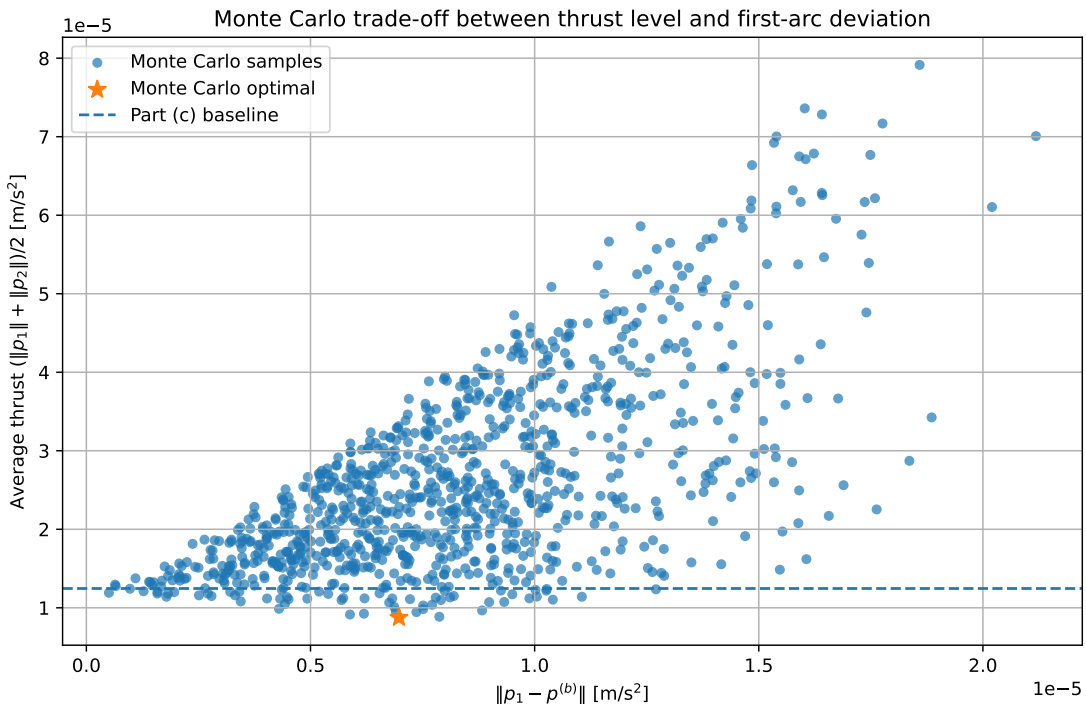

Figure 15

d.ii

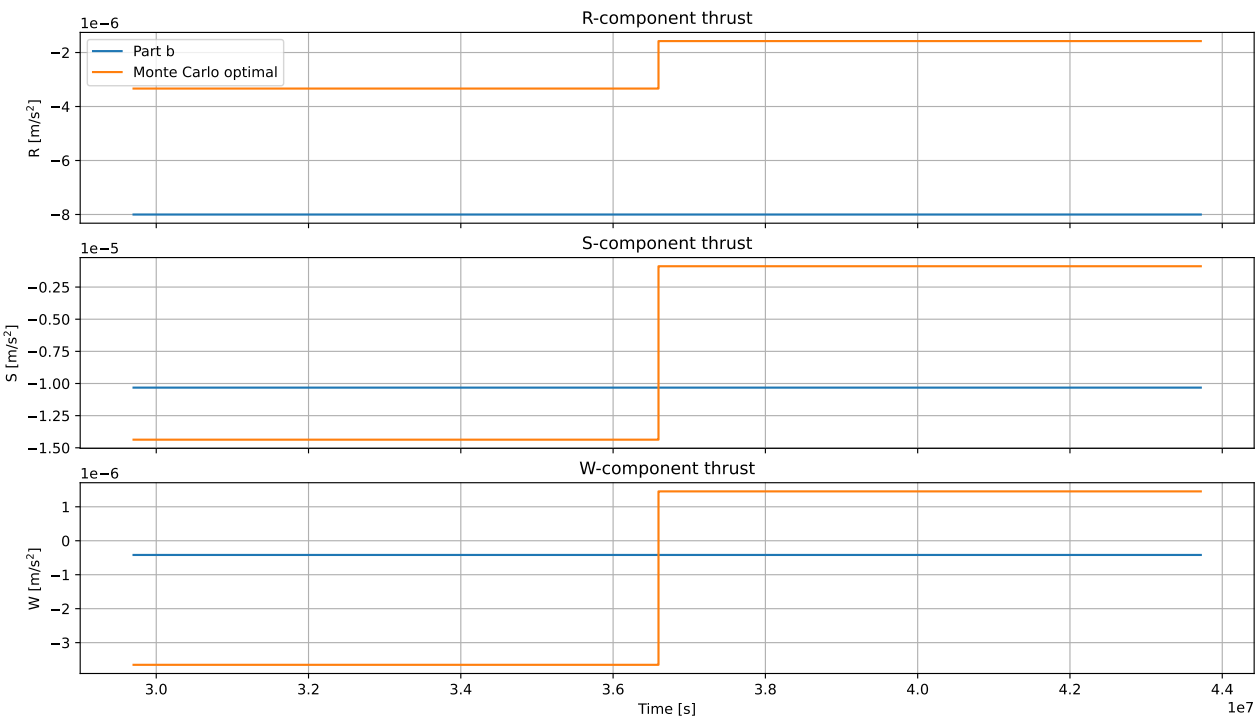

Figure 16

The Monte Carlo optimal does achieve a lower average thrust than the part (c) baseline, as the optimal sample lies below the dashed line. This is possible because fixing $\mathbf{p}_1 = \mathbf{p}^{(b)}$ is a suboptimal constraint: it forces arc 1 to

correct for the full single-arc deviation regardless of what arc 2 will do and the varying conditions. By allowing $\mathbf{p}_1$ to deviate from $\mathbf{p}^{(b)}$, the two arcs can share the correction burden more efficiently.

The reason a lower average exists is rooted in the sensitivity matrix $\mathbf{S}_r$: the columns of $\mathbf{S}_r$ represent how effectively each RSW component affects the final positional deviation, and this changes over the trajectory as the RSW frame rotates with the spacecraft. A thrust direction that is poorly aligned with the required correction at one arc epoch may be far more effective at another. The optimal $\mathbf{p}_1$ found here exploits this by applying a larger W-component in arc 1 (where it is evidently efficient) while allowing arc 2 to apply a much smaller correction, thus reducing the average thrust below what either the single-arc or the constrained two-arc solution achieves.